

Escuela Politécnica Nacional

Evaluación 2 "Análisis de Circuitos Eléctricos"

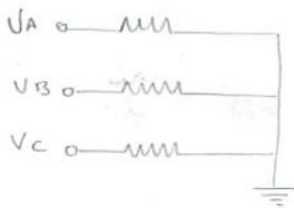
Por: Marcos Quillupangui

Resolución de la Prueba N2

- El factor de potencia de cada fase de una **carga balanceada conectada en estrella (Y)** con una impedancia de 5Ω es **0.6 en atraso** (es decir, $|Z_Y| = 5$). Esta impedancia está conectada a una fuente que tiene un voltaje **de fase de 100 V** ($V_p = 100 V_{rms}$). Determine la (a) **potencia aparente absorbida por el circuito** (1 ptos.) y (b) el diagrama fasorial indicando las corrientes y voltajes de línea, incluya el valor de los ángulos (1.5 ptos.).
Nota: para el literal b, asuma que el voltaje de línea V_{ab} tiene un ángulo de 0° , y la fuente tiene $\sec (+)$.

$$\textcircled{1} |Z_Y| = 5 [\Omega] \quad V_p = 100 [V_{rms}] \quad f_p = 0,6 (-) \quad \theta = \cos^{-1}(0,6) = 53,13^\circ$$

$$\Rightarrow Z_Y = 5 \angle 53,13 \equiv 3 + 4j [\Omega] //$$



$$a) P = 3 \cdot \frac{V_p^2}{Z} \cdot \cos \theta = 3 \cdot \frac{100^2}{5} \cdot 0,6 = 3600 [W]$$

$$\Rightarrow f_p = \frac{P}{S} \Rightarrow S = \frac{P}{f_p} = 6000 [VA] //$$

$$b) V_{ab} = |V| \angle 0^\circ \quad \sec (+)$$

$$\Rightarrow V_a = 100 \angle -30^\circ [V_{rms}] \quad V_{ab} = 100\sqrt{3} \angle 0^\circ$$

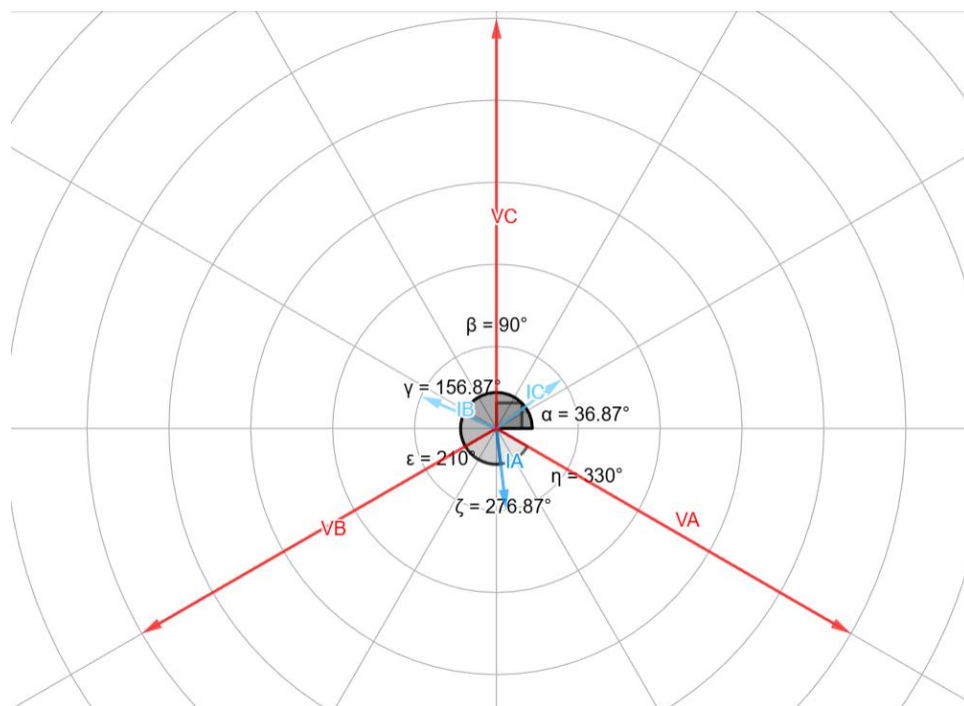
$$V_b = 100 \angle -150^\circ [V_{rms}] \quad V_{bc} = 100\sqrt{3} \angle -120^\circ$$

$$V_c = 100 \angle 90^\circ [V_{rms}] \quad V_{ca} = 100\sqrt{3} \angle 120^\circ$$

$$I_c = \frac{V_a}{Z_c} = \frac{100 \angle -30^\circ}{3 + 4j} = 20 \angle -83,13^\circ$$

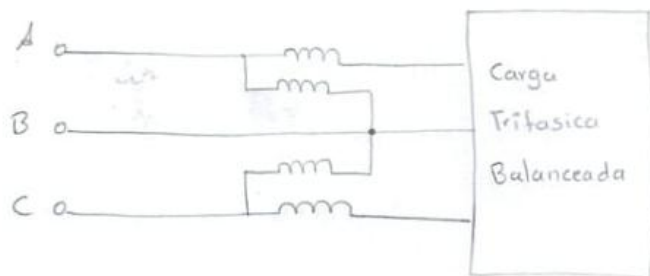
$$I_b = \frac{V_b}{Z_a} = \frac{100 \angle -150^\circ}{3 + 4j} = 20 \angle 156,87^\circ$$

$$I_c = \frac{V_c}{Z_c} = \frac{100 \angle 90^\circ}{3 + 4j} = 20 \angle 36,87^\circ$$



2. Una carga trifásica balanceada opera a $V_L = 440$ (V_{rms}) con un factor de potencia de 0.6 en atraso. Dos vatímetros se utilizan para medir la potencia de entrada, obteniéndose una lectura total de 20 (kW). Determine las lecturas individuales de los vatímetros, considerando que el punto común de conexión de los vatímetros corresponde a la **línea b**, tal como se revisó en clase (3 pts).

② $V_L = 440$ [V_{rms}] $f_p = 0,6 (-)$ $P_T = 20$ Kw $\cos \theta = 0,6 \Rightarrow \theta = 53,13^\circ$
 $\Rightarrow P_T = P_{w1} + P_{w2} \Rightarrow P_{w1} + P_{w2} = 20 \text{ Kw} //$ $\Rightarrow P_{w1} = I_a \cdot V_{ab} \cdot \cos(\theta_V - \theta_i)$
 $P_{w2} = I_c \cdot V_{cb} \cdot \cos(\theta_V - \theta_i)$



$\Rightarrow P_{fuse} = 6,6$ Kw

$\Rightarrow f_p = \frac{P}{S} \Rightarrow S = \frac{P}{f_p} = \frac{6,6 \times 10^3}{0,6} = 11,11$ K [VA]

$|S_F| = \frac{|I_F|}{|V_F|} = |I_F| = \frac{|S_F|}{|V_F|} = \frac{11,11 \times 10^3}{440} = 25,25$ [A]

$\Rightarrow V_a = 440 \angle 0^\circ \quad | \quad I_a = 25,25 \angle -53,13^\circ \quad | \quad V_{ab} = 440\sqrt{3} \angle 30^\circ$
 $V_b = 440 \angle -120^\circ \quad | \quad I_b = 25,25 \angle -173,13^\circ \quad | \quad V_{bc} = 440\sqrt{3} \angle -90^\circ$
 $V_c = 440 \angle 120^\circ \quad | \quad I_c = 25,25 \angle -66,87^\circ \quad | \quad V_{cc} = 440\sqrt{3} \angle 150^\circ$

$\Rightarrow P_{w1} = I_a \cdot V_{ab} \cdot \cos(\theta_V - \theta_i)$

$P_{w1} = 25,25 \cdot 440\sqrt{3} \cdot \cos(30 + 53,13) = 2301,8$ (w)

$P_{w2} = I_c \cdot V_{cb} \cdot \cos(\theta_V - \theta_i)$

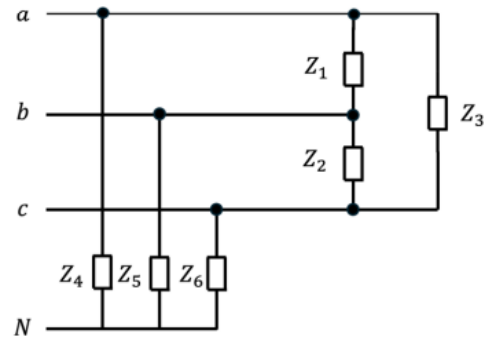
$P_{w2} = 25,25 \cdot 440\sqrt{3} \cdot \cos(+90 - 66,87) = 17696,247$ (w) //

$\Rightarrow P_{w1} + P_{w2} = 17696,247 + 2301,8 = 19998,047$ (w) //

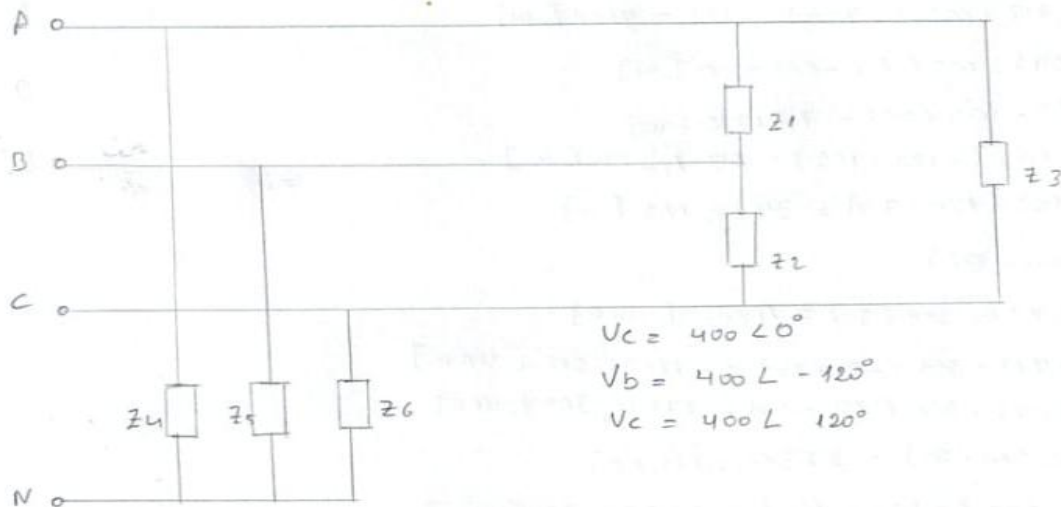
3. Una carga conectada en **estrella (Y)** y una carga conectada en **delta (Δ)** son alimentadas por un sistema trifásico de **cuatro hilos, 400 V, 50 Hz**.

- Las conexiones para la carga en **estrella** son: $2\angle 20^\circ (\Omega)$ para la línea a; $6\angle 75^\circ (\Omega)$ para la línea b; y $30\angle 50^\circ (\Omega)$ para la línea c.
- Las conexiones para la carga en **delta** son: $50\angle 30^\circ (\Omega)$ entre la línea a y b; $25\angle -60^\circ (\Omega)$ entre la línea b y c; $17.3\angle 90^\circ (\Omega)$ entre la línea a y c.

Determine (a) La potencia activa en la carga en delta; (b) La potencia reactiva en la carga en Y; (c) la potencia aparente total del sistema.



③ $V_F = 400 \text{ [V]} \quad f = 50 \text{ Hz}$



$$V_C = 400 \angle 0^\circ$$

$$V_B = 400 \angle -120^\circ$$

$$V_A = 400 \angle 120^\circ$$

$$Z_1 = 50 \angle 30^\circ = 43,301 + 25j$$

$$Z_2 = 25 \angle -60^\circ = 12,50 - 21,651j$$

$$Z_3 = 17,3 \angle 90^\circ = 17,3j$$

$$Z_4 = 2 \angle 20^\circ = 1,879 + 0,684j$$

$$Z_5 = 6 \angle 75^\circ = 1,553 + 5,796j$$

$$Z_6 = 30 \angle 50^\circ = 19,284 + 22,98j$$

$$\Rightarrow V_{AB} = V_A - V_B = 692,82 \angle 30^\circ$$

$$V_{BC} = V_B - V_C = 692,82 \angle -90^\circ$$

$$V_{CA} = V_C - V_A = 692,82 \angle 150^\circ$$

$$\Rightarrow I_{Z1} = \frac{V_{AB}}{Z_1} = 13,856 \text{ [A]}$$

$$I_{Z2} = \frac{V_{BC}}{Z_2} = \frac{692,82 \angle -90^\circ}{12,5 - 21,651j} = 24 - 13,856j \approx 27,712 \angle -30^\circ$$

$$I_{Z3} = \frac{V_{CA}}{Z_3} = \frac{692,82 \angle 150^\circ}{17,3j} = 20,024 + 34,682j \approx 40,047 \angle 60^\circ$$

$$I_{Z4} = \frac{V_A}{Z_4} = \frac{400 \angle 0^\circ}{1,879 + 0,684j} = 187,971 - 68,426j \approx 200,038 \angle -20^\circ$$

$$I_{Z5} = \frac{V_B}{Z_5} = \frac{400 \angle -120^\circ}{1,553 + 5,796j} = -64,39 + 17,254j \approx 66,662 \angle 165^\circ$$

$$I_{Z6} = \frac{V_C}{Z_6} = \frac{400 \angle 120^\circ}{19,284 + 22,98j} = 4,56 + 12,529j \approx 13,333 \angle 70^\circ$$

$$\Rightarrow P = V \cdot I \cdot \cos(\theta_v - \theta_i)$$

$$P_{z1} = 692,82 \cdot 13,856 \cdot \cos(30 - 0) = 8313,6 \text{ [W]}$$

$$P_{z2} = 692,82 \cdot 27,712 \cdot \cos(-90 - (-30)) = 9600 \text{ [W]}$$

$$P_{z3} = 692,82 \cdot 40,047 \cdot \cos(150 - 60) = 0 \text{ [W]}$$

$$P_{z4} = 400 \cdot 200,078 \cdot \cos(20) = 75,19 \text{ K [W]}$$

$$P_{z5} = 400 \cdot 66,662 \cdot \cos(-120 - 165) = 6901,358 \text{ [W]}$$

$$P_{z6} = 400 \cdot 13,333 \cdot \cos(120 - 70) = 3428,115 \text{ [W]}$$

$$\Rightarrow Q = V \cdot I \cdot \sin(\theta_v - \theta_i)$$

$$Q_{z1} = 692,82 \cdot 13,856 \cdot \sin(30) = 4800 \text{ [VAR]}$$

$$Q_{z2} = 692,82 \cdot 27,712 \cdot \sin(-90 + 30) = -16625,512 \text{ [VAR]}$$

$$Q_{z3} = 692,82 \cdot 40,047 \cdot \sin(150 - 60) = 27745,363 \text{ [VAR]}$$

$$Q_{z4} = 400 \cdot 200,078 \cdot \sin(20) = 27366,81 \text{ [VAR]}$$

$$Q_{z5} = 400 \cdot 66,661 \cdot \sin(-120 - 165) = 25756,22 \text{ [VAR]}$$

$$Q_{z6} = 400 \cdot 13,333 \cdot \sin(120 - 70) = 4085,468 \text{ [VAR]}$$

$$a) P_T(\Delta) = P_{z1} + P_{z2} + P_{z3} = 8313,6 + 9600 + 0 = 17913,6 \text{ (W) \%}$$

$$b) Q_T(y) = Q_{z4} + Q_{z5} + Q_{z6} = 27366,81 + 25756,22 + 4085,468 = 57208,498 \text{ [VA]}$$

$$c) S_{\text{Total}} = S(\Delta) + S(y)$$

$$\bullet S(\Delta) = \sqrt{P(\Delta)^2 + Q(\Delta)^2} = \sqrt{17913,6^2 + 15919,851^2} = 23965,365 \text{ [VA]}$$

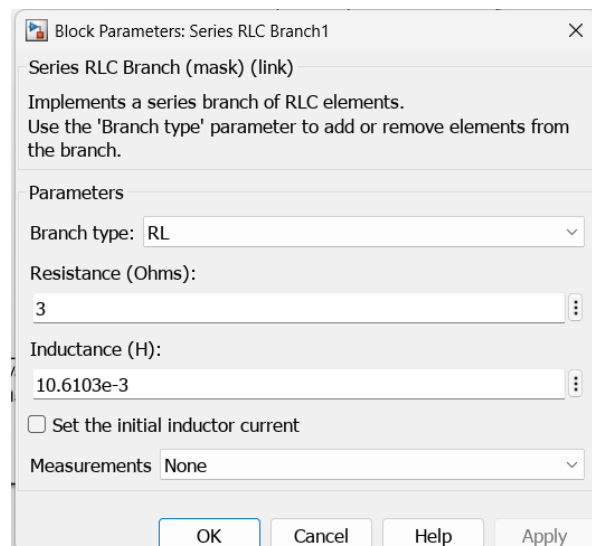
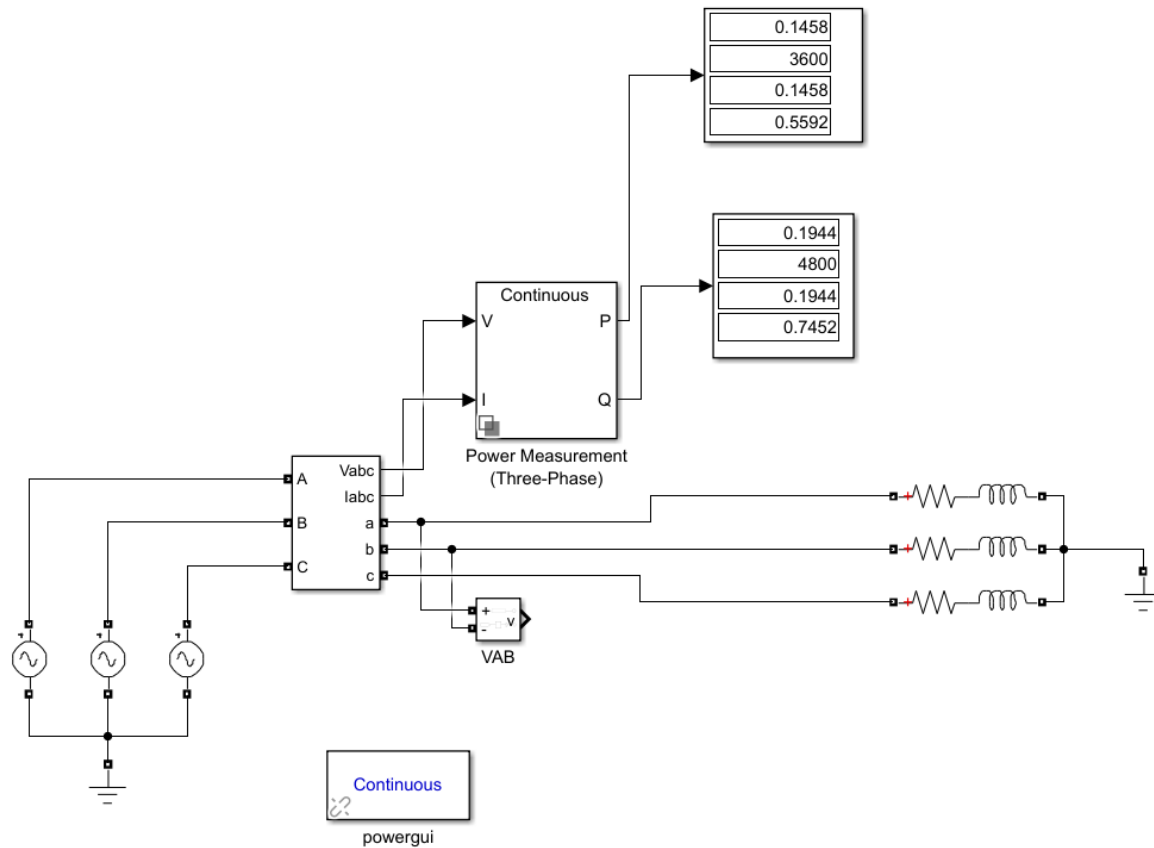
$$\bullet S(y) = \sqrt{P(y)^2 + Q(y)^2} = \sqrt{85519,473^2 + 57208,498^2} = 102892,196$$

$$\Rightarrow S_{\text{Total}} = 126855,561 \text{ [VAR] \%}$$

Circuitos Realizados en Simulink

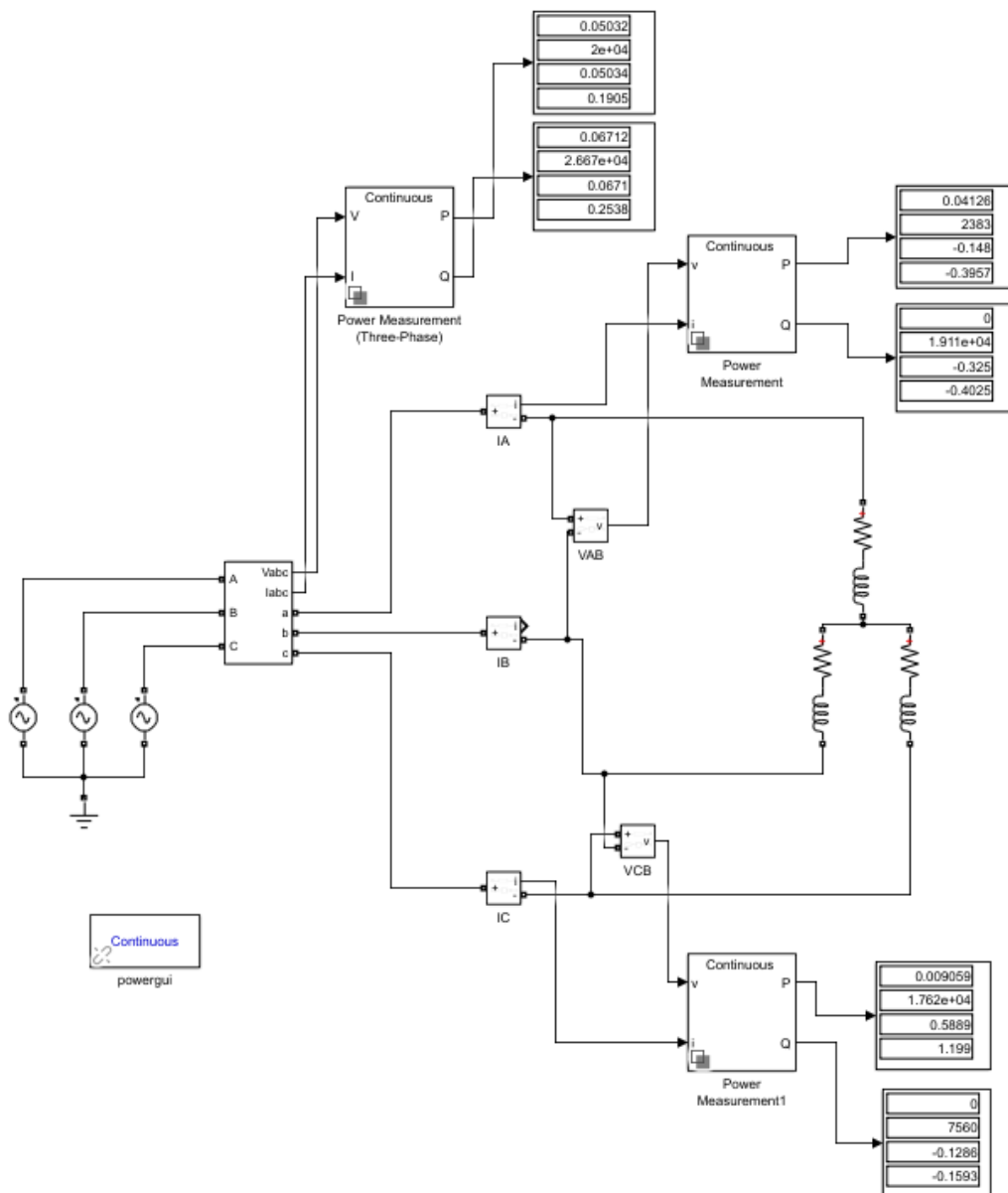
Nota: En los vatímetros, la medida a considerar es el segundo valor de la columna

Ejercicio 1

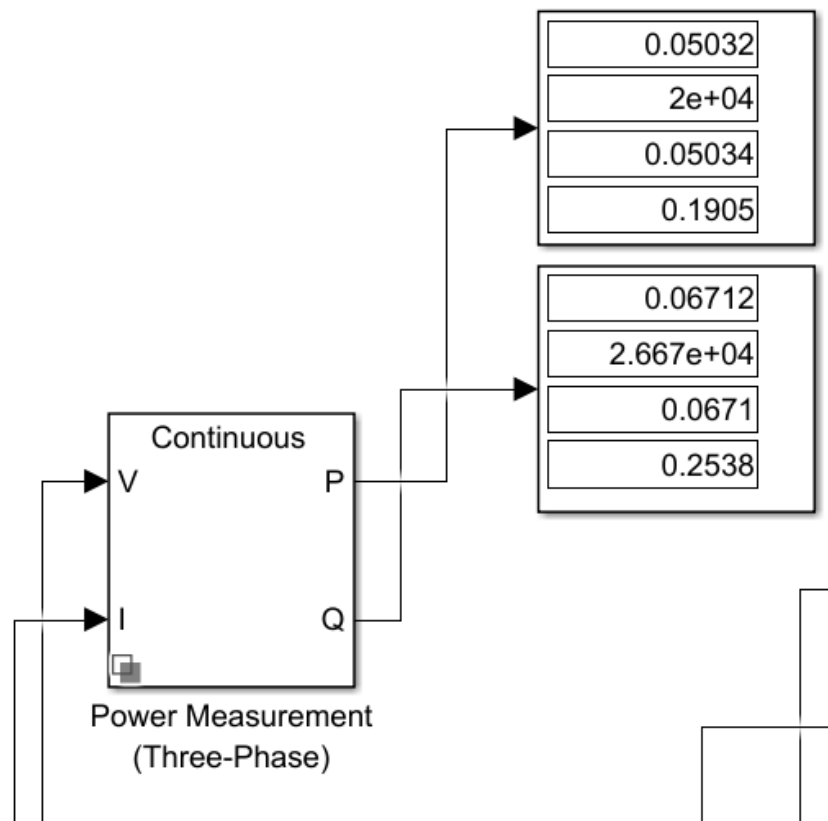


#	Type	Block	Value	Angle
MEASUREMENTS:				
1	U A	Three-Phase V-I Measurement	100 Vrms	-30.00 °
2	U B	Three-Phase V-I Measurement	100 Vrms	-150.00 °
3	U C	Three-Phase V-I Measurement	100 Vrms	90.00 °
4	U	VAB	173.2 Vrms	0.00 °
5	I A	Three-Phase V-I Measurement	20 Arms	-83.13 °
6	I B	Three-Phase V-I Measurement	20 Arms	156.87 °
7	I C	Three-Phase V-I Measurement	20 Arms	36.87 °

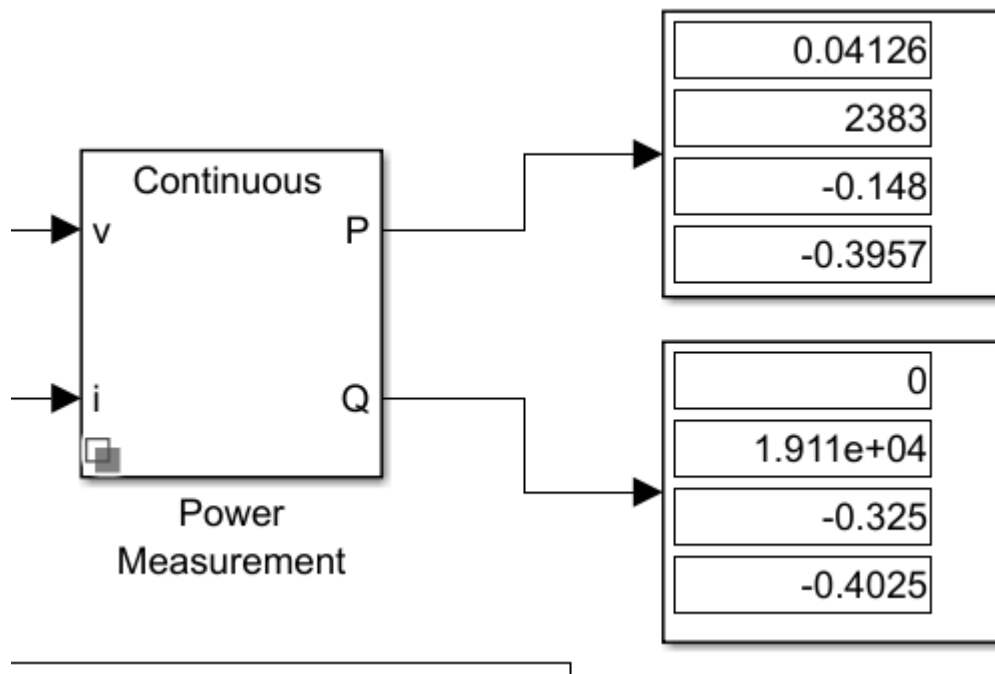
Ejercicio 2



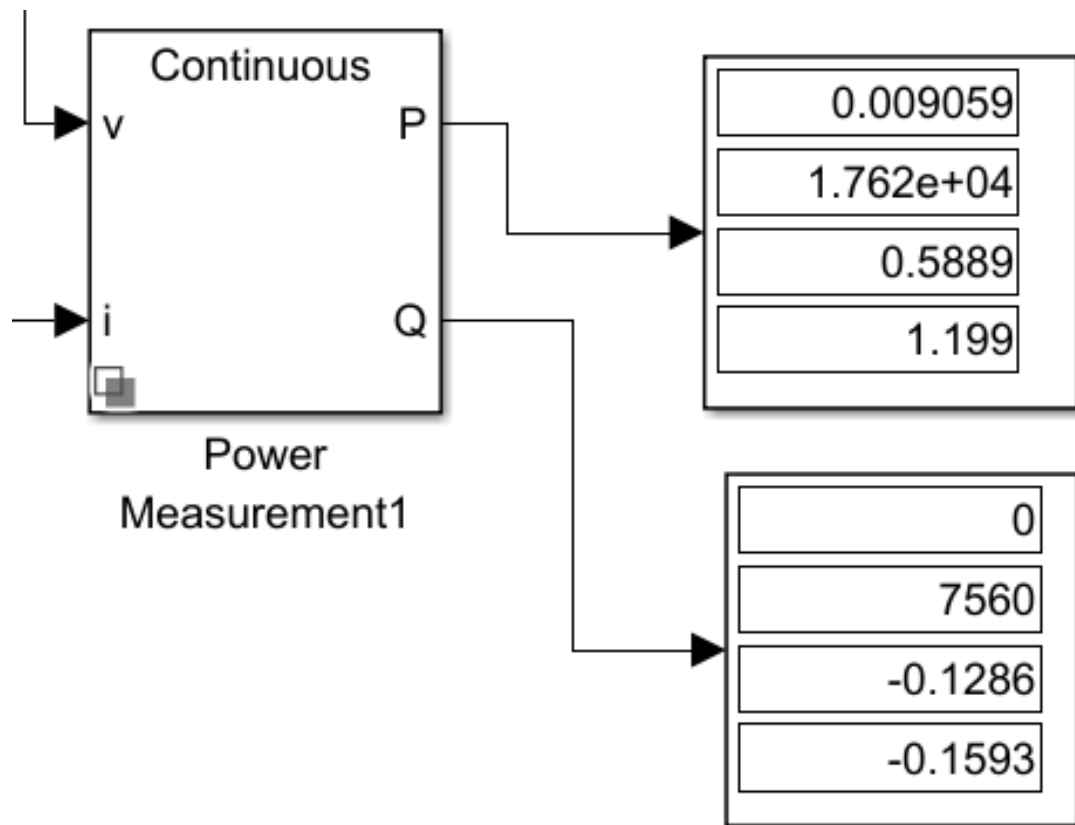
Vatímetro Total:



Vatímetro 1:



Vatímetro 2:




Block Parameters: Series RLC Branch1

Series RLC Branch (mask) (link)

Implements a series branch of RLC elements.
Use the 'Branch type' parameter to add or remove elements from the branch.

Parameters

Branch type: RL

Resistance (Ohms):
10.4544

Inductance (H):
36.975e-3

☐ Set the initial inductor current

Measurements None

OK

Cancel

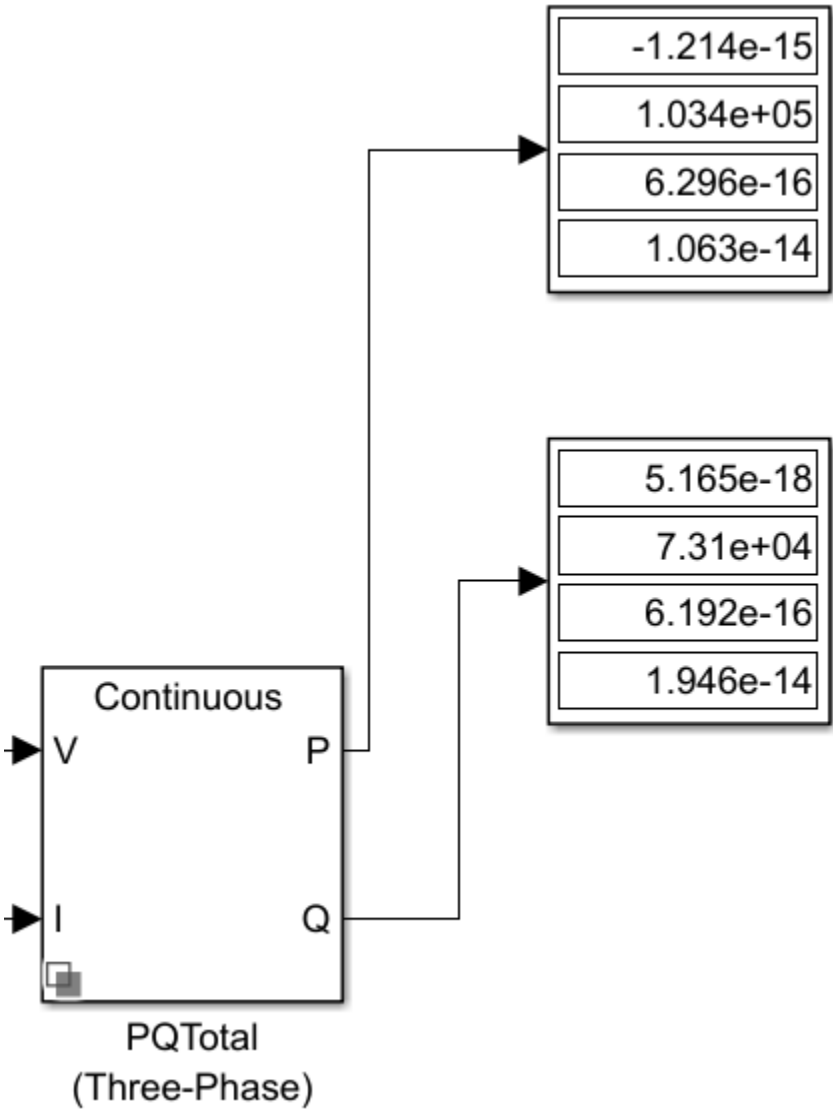
Help

Apply

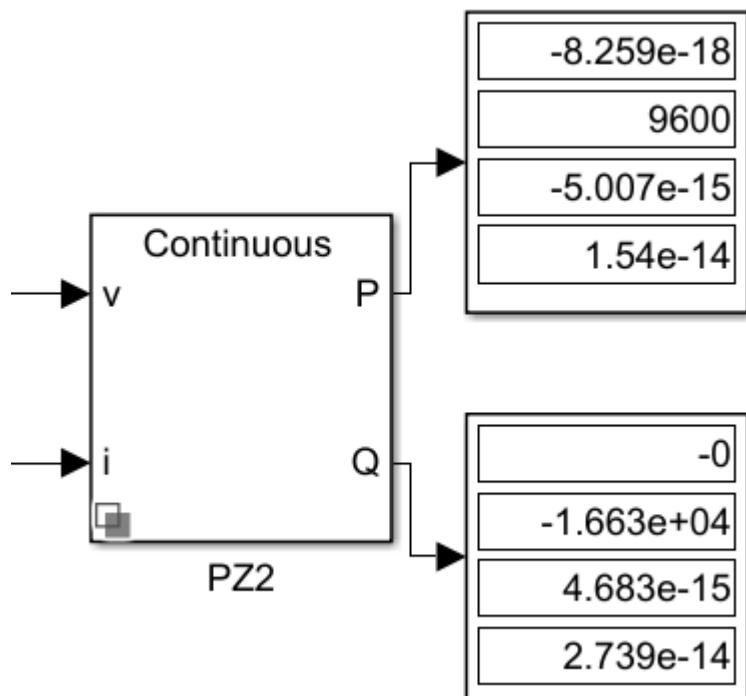
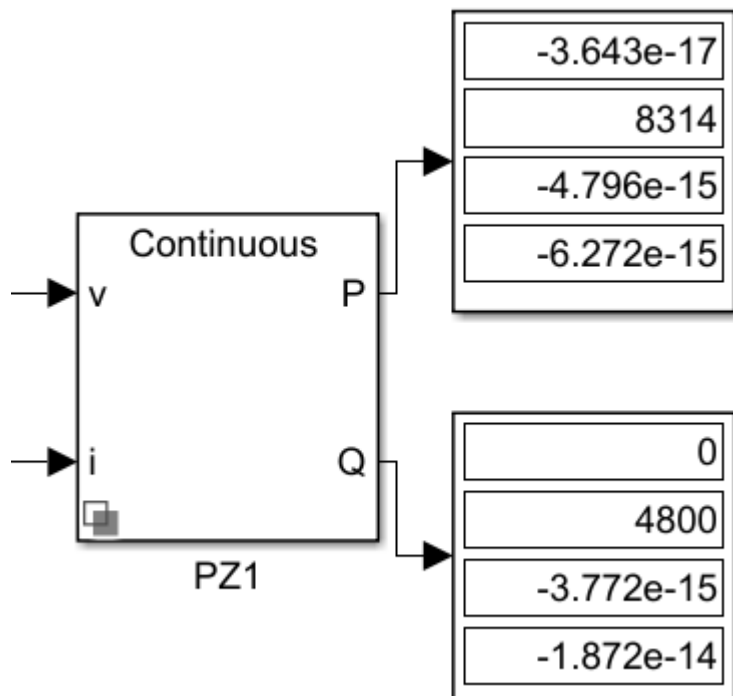
#	Type	Block	Value	Angle
	MEASUREMENTS:			
1	U A	Three-Phase V-I Measurement	440 Vrms	0.00 °
2	U B	Three-Phase V-I Measurement	440 Vrms	-120.00 °
3	U C	Three-Phase V-I Measurement	440 Vrms	120.00 °
4	U	VAB	762.1 Vrms	30.00 °
5	U	VCB	762.1 Vrms	90.00 °
6	I	IA	25.25 Arms	-53.13 °
7	I	IB	25.25 Arms	-173.13 °
8	I	IC	25.25 Arms	66.87 °
9	I A	Three-Phase V-I Measurement	25.25 Arms	-53.13 °
10	I B	Three-Phase V-I Measurement	25.25 Arms	-173.13 °
11	I C	Three-Phase V-I Measurement	25.25 Arms	66.87 °

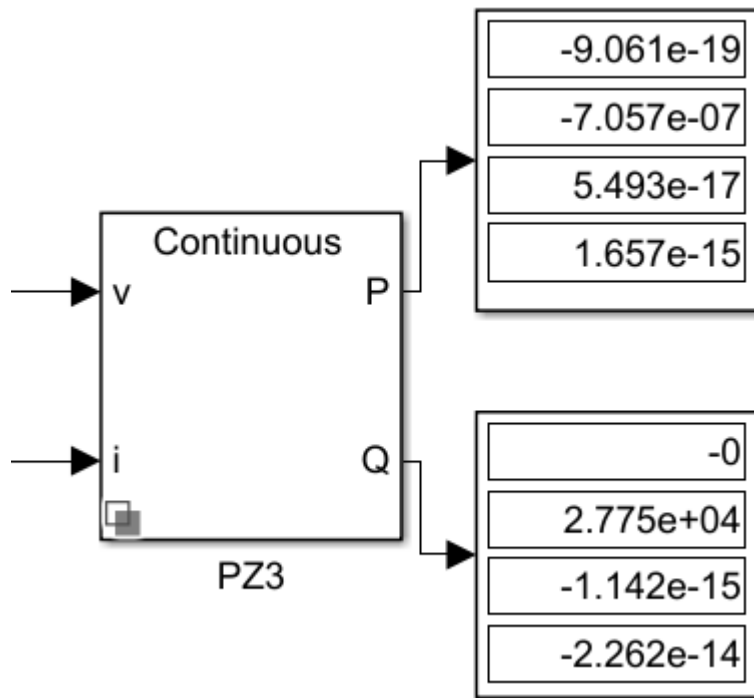
Ejercicio 3

Vatímetro Total:

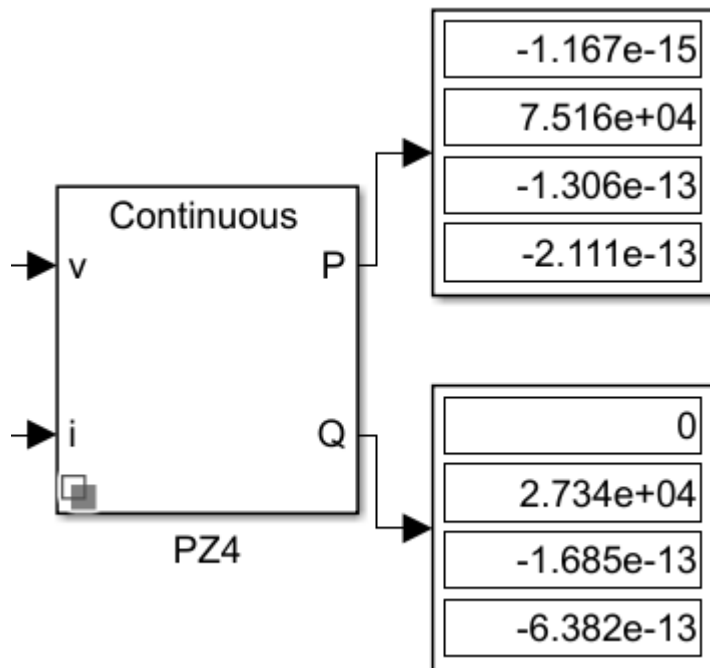


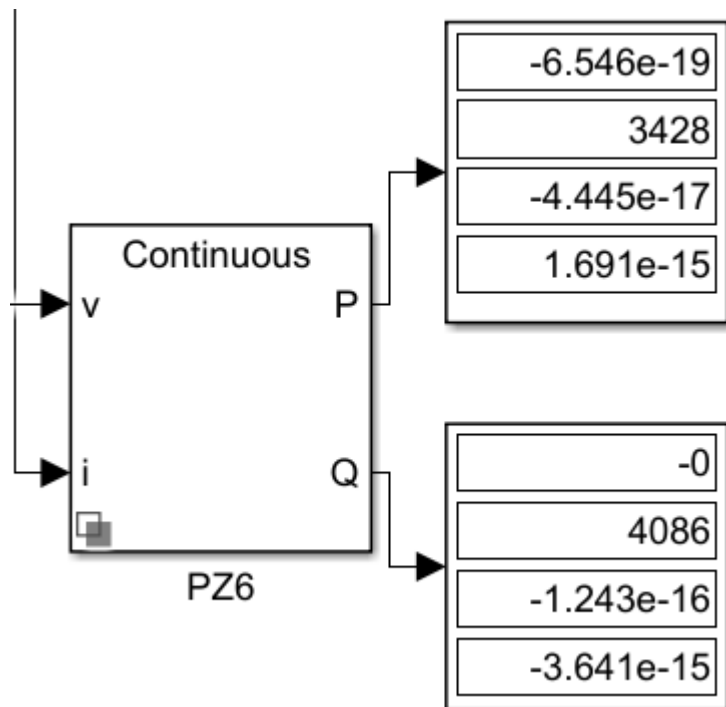
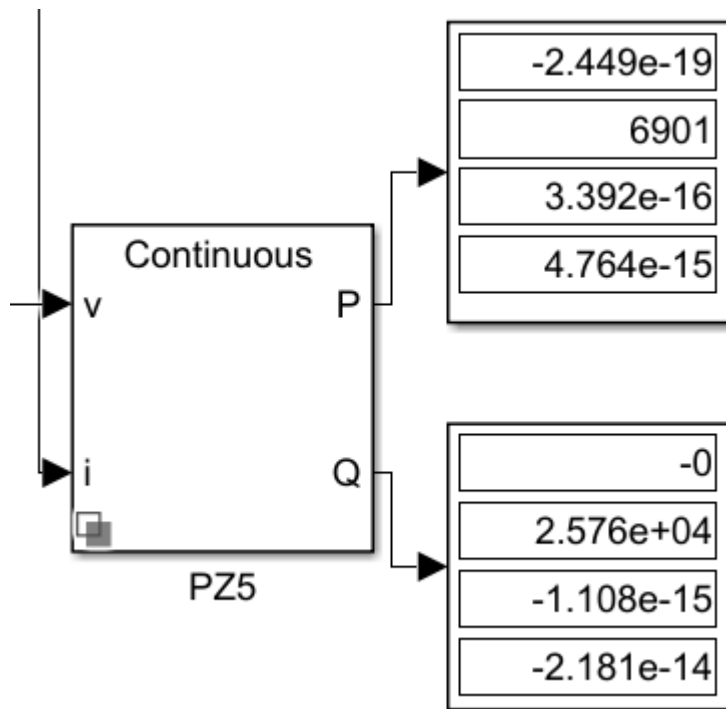
Vatímetros Delta:





Vatímetros Estrella:





#	Type	Block	Value	Angle
	MEASUREMENTS:			
1	U A	Three-Phase V-I Measurement	400 Vrms	0.00 °
2	U B	Three-Phase V-I Measurement	400 Vrms	-120.00 °
3	U C	Three-Phase V-I Measurement	400 Vrms	120.00 °
4	U	VZ1	692.8 Vrms	30.00 °
5	U	VZ2	692.8 Vrms	-90.00 °
6	U	VZ3	692.8 Vrms	150.00 °
7	U	VZ4	400 Vrms	0.00 °
8	U	VZ5	400 Vrms	-120.00 °
9	U	VZ6	400 Vrms	120.00 °
10	I	IZ1	13.86 Arms	-0.00 °
11	I	IZ2	27.71 Arms	-30.00 °
12	I	IZ3	40.05 Arms	60.00 °
13	I	IZ4	199.9 Arms	-19.99 °
14	I	IZ5	66.66 Arms	165.00 °
15	I	IZ6	13.33 Arms	70.00 °
16	I A	Three-Phase V-I Measurement	208.9 Arms	-29.55 °
17	I B	Three-Phase V-I Measurement	54.35 Arms	176.42 °
18	I C	Three-Phase V-I Measurement	61.07 Arms	89.45 °